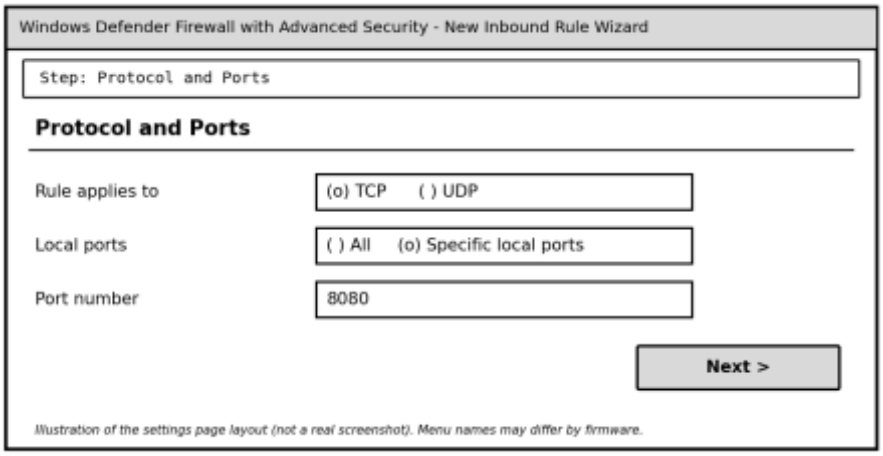
Secure Networks with Firewalls, NAT, Port Filtering and Packet Filtering

Assignment Answers

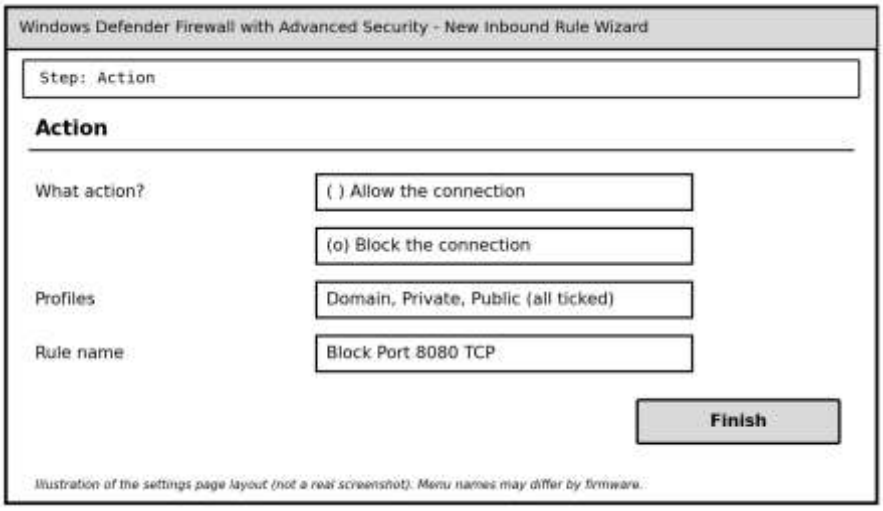
Q1. Set up Windows Defender Firewall on your Windows laptop or PC and create a new inbound rule to block all incoming traffic on port 8080. Take a screenshot of the rule you created.

Steps (graphical method)

1. Press the Windows key, type Windows Defender Firewall with Advanced Security and open it (or open Windows Security, Firewall and network protection, Advanced settings). Run it as Administrator.
2. In the left pane click Inbound Rules, then click New Rule in the right pane.
3. Select Port as the rule type and click Next.
4. Choose TCP, select Specific local ports and type 8080. Click Next.



5. Select Block the connection and click Next.
6. Tick all profiles (Domain, Private and Public) and click Next.
7. Type the name Block Port 8080 TCP and click Finish. Repeat the same steps with UDP to create Block Port 8080 UDP.



The rules in the Inbound Rules list

Windows Defender Firewall with Advanced Security - Inbound Rules

Name	Profile	Enabled	Action	Protocol	Local Port
Block Port 8080 TCP	All	Yes	Block	TCP	8080
Block Port 8080 UDP	All	Yes	Block	UDP	8080

Sample illustration of the expected result (not a real screenshot)

Same configuration with commands (Command Prompt as Administrator)

Administrator: Command Prompt _ [] X

```
C:\Windows\system32> netsh advfirewall firewall add rule name="Block Port 8080 TCP"
     dir=in action=block protocol=TCP localport=8080
Ok.

C:\Windows\system32> netsh advfirewall firewall add rule name="Block Port 8080 UDP"
     dir=in action=block protocol=UDP localport=8080
Ok.

C:\Windows\system32> netsh advfirewall firewall show rule name="Block Port 8080 TCP"

Rule Name:                 Block Port 8080 TCP
----------------------------------------------
Enabled:                   Yes
Direction:                 In
Profiles:                  Domain,Private,Public
Protocol:                  TCP
LocalPort:                 8080
RemotePort:                Any
Action:                    Block
Ok.
```

Sample illustration of the expected result (not a real screenshot)

Testing the rule

- Start a small test server on this computer (for example python -m http.server 8080) and open http://[this computer's IP]:8080 from a phone or another computer.
- Before the rule the page opens. After the rule the connection fails or times out, which shows that port 8080 is blocked.

Windows PowerShell (run on a second computer) _ [] X

```
PS C:\Users\Office> Test-NetConnection -ComputerName 192.168.0.105 -Port 8080

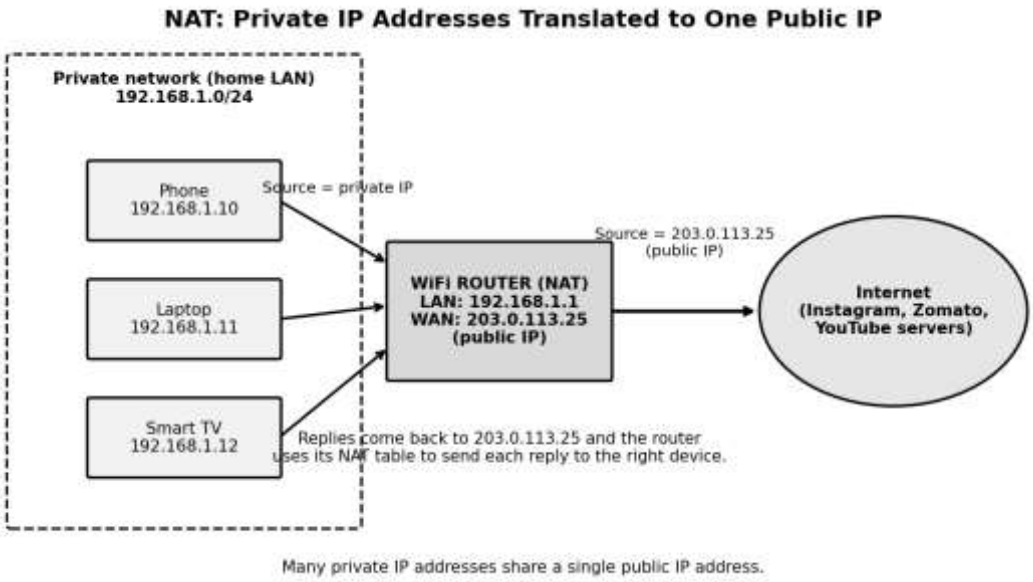
WARNING: TCP connect to (192.168.0.105 : 8080) failed

ComputerName     : 192.168.0.105
RemoteAddress    : 192.168.0.105
RemotePort       : 8080
InterfaceAlias   : Wi-Fi
SourceAddress    : 192.168.0.102
TcpTestSucceeded : False
```

Sample illustration of the expected result (not a real screenshot)

Q2. Simulate NAT (Network Address Translation) by drawing a simple diagram showing how your home WiFi router assigns private IP addresses to your phone, laptop, and smart TV, and how it translates them to a single public IP for internet access.

Hint: Use any drawing tool or even hand-draw and click a photo.



How NAT works

1. The router uses DHCP to give each device in the home a private IP address: phone 192.168.1.10, laptop 192.168.1.11 and smart TV 192.168.1.12. Private addresses (192.168.x.x, 10.x.x.x, 172.16-31.x.x) are not routable on the internet.
2. The ISP gives the router only one public IP address on its WAN side (here 203.0.113.25 as an example).
3. When the phone opens Instagram, the router changes the source address of the packet from the private IP and port (192.168.1.10:51001) to its own public IP and a new port (203.0.113.25:40001) and saves this in its NAT table.
4. The reply from the server comes to the public IP and port 40001. The router checks the NAT table and sends it back to 192.168.1.10:51001. In the same way the laptop and the smart TV use different port numbers on the same public IP.

Router NAT (PAT) Translation Table (example)

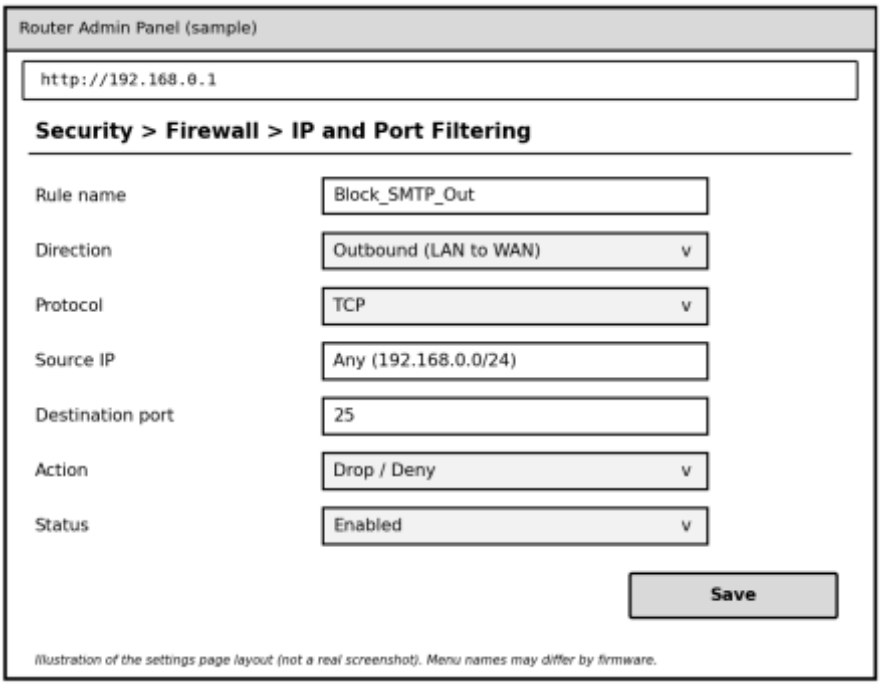
Device	Private IP : port	Public IP : port	Destination
Phone	192.168.1.10 : 51001	203.0.113.25 : 40001	Instagram server : 443
Laptop	192.168.1.11 : 52002	203.0.113.25 : 40002	Zomato server : 443
Smart TV	192.168.1.12 : 53003	203.0.113.25 : 40003	YouTube server : 443

Example illustration (made-up addresses)

Benefits: many devices share one public address (saving IPv4 addresses), and the devices are hidden from the internet because unsolicited incoming traffic has no entry in the NAT table, which gives a basic level of protection.

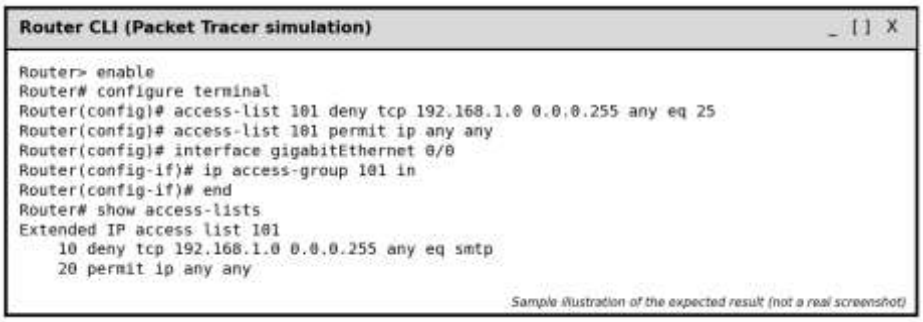
Q3. Configure port filtering using your WiFi router's admin panel (or use a router simulator online if you don't have access). Block outgoing traffic on port 25 (SMTP) and explain in 2 sentences how this could help prevent spam from your network.**Method A: router admin panel**

1. Open the router's admin page (for example http://192.168.0.1) and log in.
2. Go to Security, Firewall, IP and Port Filtering (the menu name differs between router brands; some home routers do not have outbound port filtering, and then a simulator can be used).
3. Add a new rule: Direction = Outbound (LAN to WAN), Protocol = TCP, Source = Any, Destination port = 25, Action = Drop (Deny), Status = Enabled. Save the rule.



Method B: router simulator (Cisco Packet Tracer, access list)

The access list denies TCP traffic with destination port 25 coming from the LAN and permits all other traffic. It is applied inbound on the LAN-facing interface, so it filters the traffic leaving the network.



Testing

A connection test to a mail server on port 25 now fails, while normal web browsing still works.



How this helps prevent spam

Computers infected with malware often send spam by connecting directly to mail servers on port 25, so blocking outgoing port 25 stops infected devices from sending bulk spam and protects the network's IP address from being blacklisted. Normal email programs use authenticated submission ports (587 or 465) through the mail provider, so users can still send mail while spam bots are blocked.

Q4. Given a sample packet filtering rule set (provided by your trainer), identify which of these three sample packets would be allowed or blocked: 1) HTTP request

from 192.168.1.5 to 172.217.166.78:80, 2) SSH request from 10.0.0.10 to 34.201.23.1:22, 3) DNS query from 192.168.1.7 to 8.8.8.8:53.

Hint: Focus on source/destination IPs and ports in the rule set.

The trainer's rule set was not available to me, so the sample rule set below is assumed for the working. If the trainer's rules are different, apply the same method: check the packet against the rules from top to bottom, the first matching rule decides, and a packet that matches no rule is blocked by the default deny rule.

Sample Packet Filtering Rule Set (assumed, in order from top to bottom)

Rule	Action	Protocol	Source	Destination	Dest. Port
1	Allow	TCP	192.168.1.0/24	Any	80
2	Allow	UDP	192.168.1.0/24	Any	53
3	Block	TCP	Any	Any	22
4	Block	Any	Any	Any	Any (default deny)

Assumed sample rule set for illustration (replace with the trainer's rules)

Packet	Matching rule	Decision	Reason
1) HTTP request from 192.168.1.5 to 172.217.166.78:80	Rule 1	ALLOWED	TCP, source 192.168.1.5 is inside 192.168.1.0/24 and the destination port is 80 (HTTP).
2) SSH request from 10.0.0.10 to 34.201.23.1:22	Rule 3	BLOCKED	The source 10.0.0.10 is outside 192.168.1.0/24, so rules 1 and 2 do not match, and rule 3 blocks TCP traffic to port 22 (SSH).
3) DNS query from 192.168.1.7 to 8.8.8.8:53	Rule 2	ALLOWED	UDP, source 192.168.1.7 is inside 192.168.1.0/24 and the destination port is 53 (DNS).

Q5Research and list 3 ways Instagram or Zomato might use firewalls and packet filtering to protect their servers from attacks or misuse..

1. Allowing only the needed ports (packet filtering and segmentation): A firewall at the edge of the data centre lets only HTTPS (port 443) reach the public web servers and drops everything else. Database and admin ports (such as 3306 or 22) are blocked from the internet and open only for internal application servers, so an attacker cannot connect directly to the databases that hold user data, orders or payments.
2. Rate limiting and DDoS filtering: Firewalls and traffic filters watch the number of packets and requests coming from each IP address. They drop floods such as SYN floods and block IP addresses that make too many login, OTP or order requests, which protects the servers from DDoS attacks, bots and credential-stuffing or fake-order abuse.
3. Application-layer firewall (WAF) and IP reputation filtering: A web application firewall inspects the HTTP requests and blocks suspicious patterns such as SQL injection or cross-site scripting. Packets from known malicious IP ranges or from regions where no customers exist can also be blocked, which reduces scraping and attacks on the app's APIs.